

LESSON 7-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Integers and Opposites**

Maryland Standard 6.NS.C.5 · Learning target – I can write an integer for a real situation and name its opposite.

REMEMBER

- **Integers:** ... -3 , -2 , -1 , 0 , 1 , 2 , 3 ...
- **Positive** = above 0 (gain, up). **Negative** = below 0 (loss, down).
- **Opposites** are the same distance from 0 on different sides: 6 and -6 .

WORKED EXAMPLE

A diver swims 18 feet below sea level. Write an integer for her depth and its opposite.

Step 1 Below sea level is less than 0: -18 .

Step 2 The opposite is 18 units from 0 on the other side: 18.

Answer -18 feet; opposite 18

WORD BANK

integer – ..., -2 , -1 , 0 , 1 , 2 , ...

opposite – same distance from 0, other side

negative – less than 0: below, loss, owe

positive – more than 0: above, gain, earn

PRACTICE*show your work · label units*

1. Write an integer for each situation.

Step 1 Is it up, a gain, or money in? Use a positive number.

Step 2 Is it down, a loss, or money out? Use a negative number.

a. a loss of 12 yards _____

b. a deposit of \$40 _____

c. 9 floors below the street _____

d. 15 degrees above zero _____

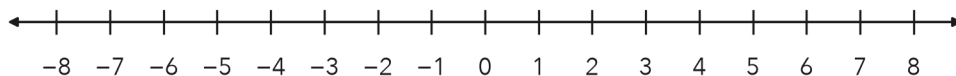
e. a \$25 withdrawal _____

f. a gain of 7 pounds _____

2. Plot -6 and its opposite on the number line. Label each point with its value.

Step 1 Find -6 on the number line and mark it.

Step 2 Its opposite is the same distance from 0, on the other side.

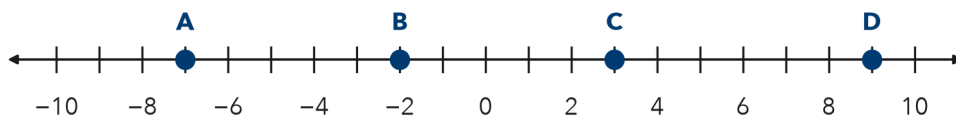


How far is each point from 0? _____ units

3. Write the integer for each point. Then write its opposite.

Step 1 Read the number under each letter.

Step 2 For the opposite, keep the number and change the sign.



Point	A	B	C	D
Integer				
Opposite				

4. **Find the Error.** Keisha says the opposite of -15 is -15 , because opposites keep the same number. Explain her mistake. Write the correct opposite.

Frame Keisha is wrong because opposites are on ___ sides of 0. The opposite of -15 is ___.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #___

AT HOME With someone at home, name a temperature on a weather report. Then name its opposite.

LESSON 7-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Integers in Real Situations**

Maryland Standard 6.NS.C.5 · Learning target – I can interpret integers in real situations and find the opposite of an opposite.

REMEMBER

- Positive: **above, gain, deposit**. Negative: **below, loss, withdrawal**.
- **0** is the start: sea level, a \$0 balance.
- The **opposite of the opposite** is the number itself:
 $-(-8) = 8$.

WORKED EXAMPLE

Find the opposite of the opposite of 14.

Step 1 The opposite of 14 is -14 .

Step 2 The opposite of -14 is 14.

Answer 14

WORD BANK

opposite – same distance from 0, other side
 $-(-9)$ – the opposite of -9 , which is 9

below zero – a negative number

PRACTICE

show your work · label units

1. Write an integer for each situation.

a. a town 35 feet below sea level ____ **b.** earning \$20 walking dogs ____

c. 4 degrees below zero ____ **d.** a loss of 8 yards ____

2. Find each value.

Step 1 Read $-(-9)$ as “the opposite of -9 .”

Step 2 Flip to the other side of 0.

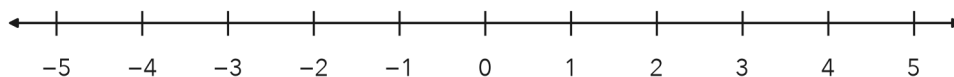
a. $-(-9) =$ ____ **b.** $-(15) =$ ____ **c.** $-(-(-4)) =$ ____

3. Start at -3 . Plot point **A** at -3 , point **B** at the opposite of A, and point **C** at the opposite of B.

Step 1 Mark A at -3 .

Step 2 B is the same distance from 0 on the other side.

Step 3 C is the opposite of B.



What do you notice about points A and C? _____

4. **Weather.** At noon it was 6 degrees **above** zero. At midnight it was 6 degrees **below** zero.
- a. Write an integer for each temperature. _____
- b. How are the two integers related? _____
5. **Reasoning.** Marcus says the opposite of a number is always negative. Is he right? Give an example.

Frame Marcus is _____. For example, the opposite of _____ is _____, which is _____.

TEST PRACTICE

choose the best answer

6. Which integer describes a loss of 18 points in a game?
- A** 18 **B** -18 **C** $-(-18)$ **D** 0
7. Which equation is true?
- A** $-(-11) = -11$ **B** $-(-11) = 11$ **C** $-(11) = 11$ **D** $-(0) = 1$

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Tell someone at home a money story with a negative number. Ask what 0 means in it.